

Unit 6: Algebraic Relationships Day 1: Independent and Dependent Variables

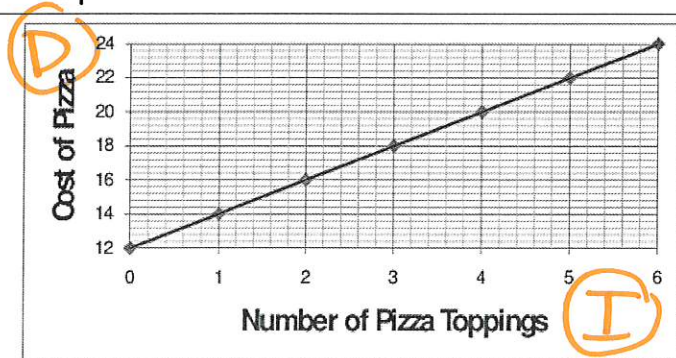
Independent Variables

- The independent variable is always located on the x axis of a graph.
- The independent variable is the input.
- It is usually the x in a table or equation.
- The independent variable STANDS ALONE
- The independent variable is what happens 1st.

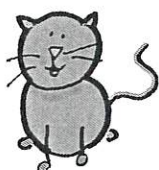
Dependent Variables

- The dependent variable is always located on the y axis of a graph.
- The dependent variable is the output.
- It is usually the y in a table or equation.
- The dependent variable DEPENDS on the independent variable.
- The dependent variable is what happens 2nd.

Examples:



Independent Variable: Pizza Toppings
 Dependent Variable: Cost of pizza
 The cost of pizza depends on # of toppings.



I
D

Number of Cats (x)	1	3	5	7
Ounces of Cat Food (y)	3	9	15	21

The table above shows the ounces of cat food needed to feed cats at the pet shelter.

Independent Variable: # of cats

Dependent Variable: ounces of food

The ounces of food depends on the number of cats

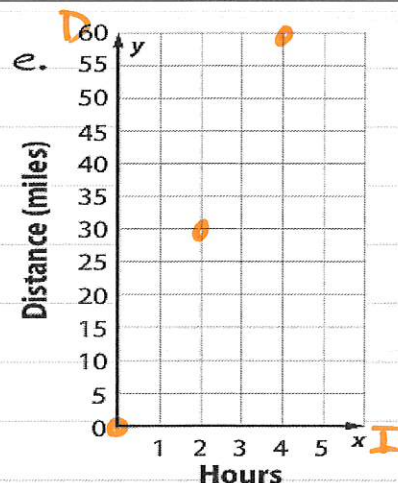
The more cats you have, the more food you need.

Briana bikes 15 miles per hour.

The rule that represents this situation is $15(x)$, where x is the number of hours. Make a table to find how many miles she has biked after 0, 2, and 4 hours. Then graph the ordered pairs.

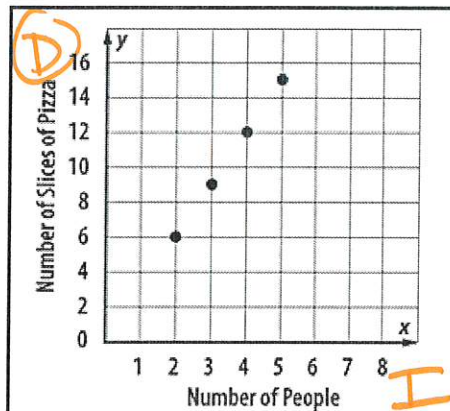
Input	Process	Output
<u>x hours</u>	<u>$15(x)$</u>	<u>y miles</u>
<u>0</u>	<u>$15(0) = 0$</u>	
<u>2</u>	<u>$15(2) = 30$</u>	
<u>4</u>	<u>$15(4) = 60$</u>	

The miles depends on hours biked.

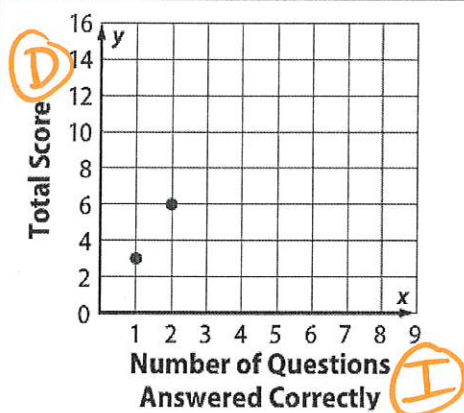


Independent Quantities: { 0, 2, 4 }

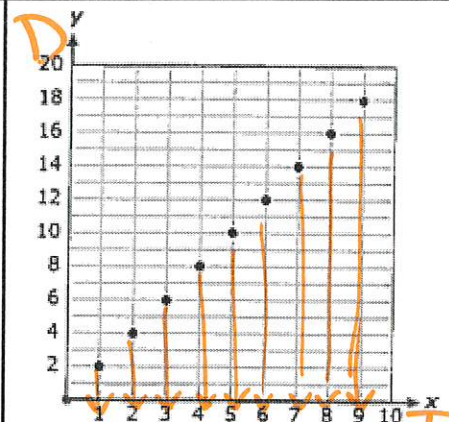
Practice: Independent and Dependent Variables



The # of slices
depends on
of people
(independent variable)



The test score
depends on
questions correct
(independent variable)



Which set represents all independent quantities shown in the graph?

- A. {1, 2, 3, 4, 5, 6, 7, 8, 9}
B. {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
C. {2, 4, 6, 8, 10, 12, 14, 16, 18}
D. {0, 2, 4, 6, 8, 10, 12, 14, 16, 18}

1 C 2 B 3 C 4 A 5 A 6 A 7 C 8 D

Which table shows an independent quantity of 3 corresponding with a dependent quantity of 6?

I D

x	y
2	4
3	5
4	6
5	7

I D

x	y
10	5
8	4
6	3
4	2

I D

x	y
2	4
3	6
4	8
5	10

I D

x	y
7	6
6	5
5	4
4	3

SIMPLIFY: $\frac{29}{8}$ $3\frac{5}{8}$ $\frac{55}{15}$ $3\frac{10}{15} = 3\frac{2}{3}$

14.5637

Round to the hundredths: 14.56

Round to the tenths: 14.6

Round to the ones: 15

Round to the thousandths: 14.564

Write about it: Describe the differences between independent and dependent variables. Be sure to write in complete sentences and use punctuation. (2 or more sentences)

The independent variable represents the input or x-axis. The dependent variable represents the output or y-axis.

